

A Course in Modern Macroeconomics

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Complete Solutions

Chapter 6 — Consumption and Saving

All nine end-of-chapter problems, 6.1–6.9 (pp. 121–125)

Every problem carries the full question text, a step-by-step solution with no skipped algebra, an economic reading of the result, and a verification. Numerical answers are reproduced by the companion scripts in `Resolucao/kurlat_ch06_codigo/`, which check each analytical formula against an independent computation — in most cases a direct numerical maximisation of the household’s problem — and abort on any mismatch.

Problems are grouped by **topic**, not by number, following the order of the chapter’s own sections. Blue boxes are odd-numbered problems, purple boxes even.

Kurlat publishes no answer key, so every result here is verified by other means: executed numerical checks (marked ✓), the Euler equation and budget constraint substituted back, limiting cases ($\sigma \rightarrow 1$, $m \rightarrow 0$, $\beta(1+r) \rightarrow 1$), and cross-book comparison with Romer (2012) ch. 8 and Jones (2020) ch. 16, both in `Livros/`.

Companion to `kurlat_solutions_ch1-4.tex`, `..._ch05.tex` and `..._ch09.tex`. Chapter 8 is outside the course syllabus.

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Chapter 6 Consumption and Saving

The Solow model assumed a saving rate. This chapter derives one. Everything in it follows from a single optimisation problem and its two implications.

The problem. A household chooses c_1 , c_2 (and savings a) to maximise $u(c_1) + \beta u(c_2)$ subject to $a = y_1 - c_1$ and $c_2 = y_2 + (1+r)a$. Collapsing the two constraints gives the *intertemporal budget constraint*

$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r}. \quad (6.2.4)$$

Implication 1: the Euler equation. The first-order conditions give

$$u'(c_1) = \beta(1+r)u'(c_2), \quad (6.2.8)$$

the single most-used equation in modern macroeconomics. It says the marginal cost of giving up a unit today equals the discounted marginal benefit of the $(1+r)$ units it buys tomorrow, and it reappears verbatim as (6.3.9), (7.4.3) and (9.1.8).

Implication 2: the permanent income hypothesis. With CRRA utility $u(c) = c^{1-\sigma}/(1-\sigma)$ the Euler equation gives $c_2 = [\beta(1+r)]^{1/\sigma} c_1$ and, substituted into (6.2.4),

$$c_1 = \frac{y_1 + \frac{1}{1+r}y_2}{\underbrace{1 + \frac{1}{1+r}[\beta(1+r)]^{\frac{1}{\sigma}}}_{\Omega}} \quad (6.2.9)$$

Consumption is proportional to the *present value of lifetime income*, not to current income. Everything the chapter says about the marginal propensity to consume, about the Keynesian consumption function, and about fiscal policy is read off this one formula.

The notation used throughout this file. It is convenient to name the denominator of (6.2.9) once:

$$\Omega \equiv 1 + \beta^{\frac{1}{\sigma}}(1+r)^{\frac{1-\sigma}{\sigma}} = 1 + \frac{1}{1+r}[\beta(1+r)]^{\frac{1}{\sigma}}, \quad W \equiv a_0 + y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r},$$

so that $c_1 = W/\Omega$. Note that $\Omega > 1$ always, that Ω is increasing in β , and that Ω is increasing, constant or decreasing in r according as $\sigma \leq 1$. That last fact drives Problems 6.1(e), 6.3 and 6.6.

Two parameters, and do not confuse them. β is *impatience*: it says how much the household cares about the future at all. σ is the inverse of the *elasticity of intertemporal substitution*, $EIS = 1/\sigma$: it says how willing the household is to let consumption differ

across dates in response to a price. A patient household (β high) saves more; a household with a low EIS (σ high) resists tilting its consumption path whatever the interest rate does.

Where the model fails, and the two repairs. The measured marginal propensity to consume out of temporary income is far above what (6.2.9) predicts (Problem 6.8 makes the gap explicit). §6.2 adds *credit constraints* (Problem 6.5) and §6.4 sketches *behavioural* departures. Naming which repair is at work is how the chapter says consumption policy should be analysed.

The Two-Period Problem and Its Comparative Statics

Problem 6.1 Two-Period Problem with Taxes and Initial Wealth

Question

Suppose a household solves the following two-period consumption-savings problem with taxes:

$$\max_{c_1, a, c_2} u(c_1) + \beta u(c_2) \quad \text{s.t.} \quad a = a_0 + y_1 - \tau_1 - c_1, \quad c_2 = y_2 - \tau_2 + (1+r)a$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, where c_1 is consumption at time 1, c_2 is consumption at time 2, y_1 is household income at time 1, y_2 is household income at time 2, τ_1 are taxes at time 1, τ_2 are taxes at time 2, and a_0 is initial wealth.

- Solve for the household's choice of c_1 , c_2 and a in closed form.
- How does c_1/y_1 depend on y_2 ? What would happen if households suddenly became optimistic about the future?
- How does c_1/y_1 depend on a_0 ? Interpret.
- How does c_1/y_1 depend on β ? Interpret.
- Suppose $y_2 = \tau_1 = \tau_2 = 0$ and compute $\partial c_1 / \partial r$. How does the answer depend on σ ? Interpret the answer. [Hint: this is a hard question, not the maths but the interpretation. Think about what σ means for the relative importance of income and substitution effects.]

Solution:

- Step 1: collapse the two constraints.** Solve the second constraint for $a = (c_2 - y_2 + \tau_2)/(1+r)$ and substitute into the first:

$$c_1 + \frac{c_2 - y_2 + \tau_2}{1+r} = a_0 + y_1 - \tau_1 \implies c_1 + \frac{c_2}{1+r} = \underbrace{a_0 + y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}}_W. \quad (6.1.1)$$

This is (6.2.4) with taxes and initial wealth. Note where each term sits: a_0 enters undiscounted because it is already available at date 1, while $y_2 - \tau_2$ is discounted.

Step 2: the Euler equation. The Lagrangian for $\max u(c_1) + \beta u(c_2)$ subject to (6.1.1) is $\mathcal{L} = u(c_1) + \beta u(c_2) + \lambda \left[W - c_1 - \frac{c_2}{1+r} \right]$, with first-order conditions

$$u'(c_1) = \lambda, \quad \beta u'(c_2) = \frac{\lambda}{1+r} \quad \implies \quad u'(c_1) = \beta(1+r)u'(c_2).$$

With $u'(c) = c^{-\sigma}$,

$$c_1^{-\sigma} = \beta(1+r)c_2^{-\sigma} \quad \implies \quad c_2 = [\beta(1+r)]^{\frac{1}{\sigma}} c_1. \quad (6.1.2)$$

Step 3: substitute back. Putting (6.1.2) into (6.1.1),

$$c_1 \left[1 + \frac{[\beta(1+r)]^{1/\sigma}}{1+r} \right] = W \quad \implies \quad c_1 = \frac{W}{\Omega}, \quad \Omega \equiv 1 + \beta^{\frac{1}{\sigma}}(1+r)^{\frac{1-\sigma}{\sigma}}.$$

Therefore

$$\boxed{c_1 = \frac{a_0 + y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}}{\Omega}} \quad \boxed{c_2 = [\beta(1+r)]^{\frac{1}{\sigma}} \frac{W}{\Omega}}$$

$$\boxed{a = a_0 + y_1 - \tau_1 - \frac{W}{\Omega}}$$

✓ In `ch06_numerico.py` this closed form is compared, for $\sigma \in \{0.5, 1, 2, 4\}$, against a direct numerical maximisation of $u(c_1) + \beta u(c_2)$ over the budget line; the two agree to 10^{-6} , and the Euler equation and the budget constraint are each substituted back and verified.

Read the formula before doing any comparative statics. It has exactly two moving parts. The numerator W is **lifetime resources**: it is where y_1 , y_2 , a_0 and the two taxes live, and *only their present value matters* — the timing of income and of taxes is irrelevant to it. The denominator Ω is **the shape of the plan**: it is where β , σ and r live, and it determines how the given resources are split between the two dates. Every part below is a question about one or the other.

(b) **Dependence on y_2 .** From the boxed formula,

$$\frac{c_1}{y_1} = \frac{W}{\Omega y_1} \quad \implies \quad \boxed{\frac{\partial}{\partial y_2} \left(\frac{c_1}{y_1} \right) = \frac{1}{\Omega(1+r)y_1} > 0}.$$

✓ Confirmed by finite differences for every σ tested.

*If households suddenly became optimistic about the future, y_2 rises in expectation, so c_1 rises **today**, with no change whatsoever in today's income. The saving rate*

$s = 1 - c_1/y_1$ falls. This is the sharpest break with the Keynesian consumption function: consumption can boom on news alone. It is the reason consumer-confidence indices are watched, and the reason a credible announcement of future tax cuts, or of a future productivity boom, moves spending immediately — exactly the mechanism that Problem 9.9 and the anticipated-technology experiment of §9.3 run in general equilibrium.

(c) **Dependence on a_0 .**

$$\frac{\partial}{\partial a_0} \left(\frac{c_1}{y_1} \right) = \frac{1}{\Omega y_1} > 0$$

✓ Note this derivative is larger than the one in part (b) by a factor $(1+r)$, because initial wealth is available now and future income has to be discounted.

Interpretation. The ratio of consumption to current income is **not a preference parameter**. Two households with identical β , σ and current income will have different saving rates if their balance sheets differ. Three consequences:

- Cross-sectional saving rates cannot be read as evidence about thrift. The elderly have accumulated a_0 and so consume a high fraction of current income; the young have not.
- There is a **wealth effect** on consumption: a rise in house or equity prices raises a_0 and therefore raises consumption immediately, without any change in income. This is one of the main channels through which asset markets reach the real economy.
- Aggregate saving rates fall when aggregate wealth rises, which is one candidate explanation for the long decline of measured US household saving.

(d) **Dependence on β .** β appears only in Ω :

$$\frac{\partial \Omega}{\partial \beta} = \frac{1}{\sigma} \beta^{\frac{1}{\sigma}-1} (1+r)^{\frac{1-\sigma}{\sigma}} > 0 \quad \Rightarrow \quad \frac{\partial}{\partial \beta} \left(\frac{c_1}{y_1} \right) = -\frac{W}{\Omega^2 y_1} \frac{\partial \Omega}{\partial \beta} < 0$$

✓

Interpretation. More patience means less consumption today and more saving, for any σ and any r . This is the one comparative static in the whole problem with an unambiguous sign, and the reason is structural: β does not appear in W at all, so there is no offsetting wealth effect — patience is a pure preference over the shape of the path.

(e) **Dependence on r , with $y_2 = \tau_1 = \tau_2 = 0$.** Now $W = a_0 + y_1$ does not depend on r , so all the action is in Ω :

$$\frac{d\Omega}{dr} = \beta^{\frac{1}{\sigma}} \frac{1-\sigma}{\sigma} (1+r)^{\frac{1-2\sigma}{\sigma}},$$

$$\frac{dc_1}{dr} = -\frac{a_0 + y_1}{\Omega^2} \frac{d\Omega}{dr} = -(a_0 + y_1) \beta^{\frac{1}{\sigma}} \frac{1 - \sigma}{\sigma} \frac{(1 + r)^{\frac{1-2\sigma}{\sigma}}}{\Omega^2}$$

Every factor is positive except $(1 - \sigma)$, so

$$\text{sign} \frac{dc_1}{dr} = \text{sign}(\sigma - 1),$$

that is,

$$\sigma < 1 \Rightarrow c_1 \text{ falls}, \quad \sigma = 1 \Rightarrow c_1 \text{ unchanged}, \quad \sigma > 1 \Rightarrow c_1 \text{ rises}.$$

✓ Verified by finite differences: with $y_1 = 1$, $a_0 = 0.2$, $\beta = 0.96$, $r = 0.05$, $\frac{dc_1}{dr}$ equals -0.2856 , 0 , $+0.1428$ and $+0.2142$ for $\sigma = 0.5, 1, 2, 4$.

Interpretation — and the hint is right that this is the hard part. The household here has all its income today and none tomorrow, so it must be a saver. A rise in r is a **fall in the price of future consumption**, $p_2 = 1/(1 + r)$. Standard consumer theory splits the response into two:

- **Substitution effect.** Future consumption is cheaper, so tilt the plan towards it: c_1 down. Its strength is the elasticity of substitution between c_1 and c_2 , which for CRRA is exactly $\text{EIS} = 1/\sigma$.
- **Income effect.** At the old plan the household is better off — its savings now buy more — so it wants more of both goods: c_1 up.

The two are exactly balanced when the elasticity of substitution is one, which is $\sigma = 1$, the logarithmic case; then $\Omega = 1 + \beta$ and $c_1 = (a_0 + y_1)/(1 + \beta)$, with r nowhere in sight. Away from that knife edge:

- $\sigma < 1$ (**high EIS**, the household is happy to let consumption differ across dates): substitution wins and saving rises with r . This is the intuitive case and the one most people expect.
- $\sigma > 1$ (**low EIS**, the household badly wants a smooth path): income wins and saving *falls* when r rises. The intuition is the “target saver”: someone determined to reach a given standard of living tomorrow needs to set aside less today when the return is higher, and spends the difference.

This is why the sign of the aggregate saving response to interest rates is a genuinely open empirical question rather than a theoretical one, and why estimates of the EIS occupy a whole literature. Most microeconomic estimates put $1/\sigma$ below one, i.e. $\sigma > 1$ — the counterintuitive case. It is also why the sign of Ω ’s response to r is flagged in the chapter overview: it drives Problems 6.3 and 6.6 as well.

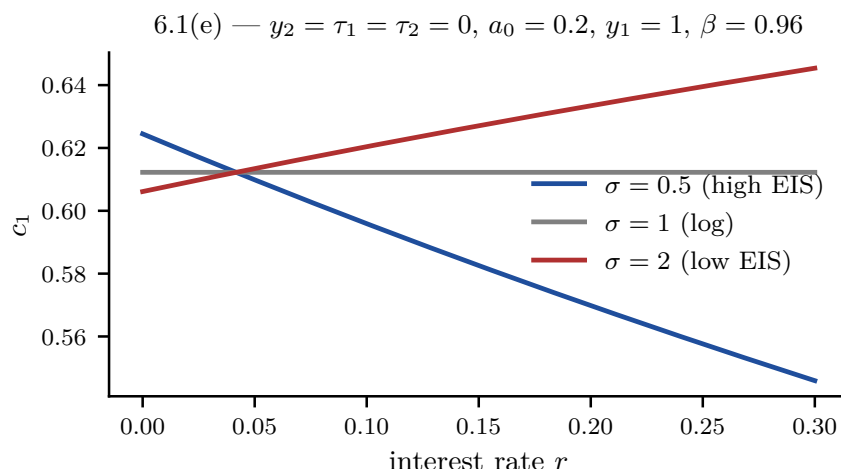


Figure 1: Problem 6.1(e). With all income in period 1, the response of c_1 to the interest rate has the sign of $\sigma - 1$. The logarithmic case is exactly flat.

Problem 6.9 The Marginal Propensity to Consume in Proportional Terms

Question

A household solves a special case of the problem in Exercise 6.1, with $\tau_1 = \tau_2 = a_0 = 0$ and $y_2 = my_1$, where m is some number.

- (a) Suppose that both y_1 and y_2 increase by $x\%$. By what percentage do c_1 and c_2 increase?
- (b) Suppose y_1 increases by $x\%$ but y_2 remains unchanged. By what percentage does c_1 increase? How does this depend on m and r ? Explain.

Setup. From Problem 6.1 with $\tau_1 = \tau_2 = a_0 = 0$ and $y_2 = my_1$,

$$W = y_1 + \frac{my_1}{1+r} = y_1 \left(1 + \frac{m}{1+r}\right) = y_1 \frac{1+r+m}{1+r}, \quad c_1 = \frac{W}{\Omega}, \quad c_2 = [\beta(1+r)]^{\frac{1}{\sigma}} c_1.$$

Solution:

- (a) If y_1 and y_2 both rise by $x\%$, then W rises by exactly $x\%$. Since c_1 is *linear* in W and Ω does not depend on income at all,

$$c_1 \text{ and } c_2 \text{ each rise by exactly } x\%$$

✓ and the saving rate $s = 1 - c_1/y_1$ is unchanged.

This is **homotheticity**. CRRA utility has the property that scaling all income scales all consumption in the same proportion, which is why the model can be aggregated and why the Kaldor fact of a stable long-run saving rate is not in conflict with optimising behaviour.

(b) Now only y_1 rises, by $x\%$. New lifetime resources:

$$W' = y_1(1+x) + \frac{my_1}{1+r} = W + x y_1,$$

so

$$\frac{W'}{W} = 1 + x \frac{y_1}{W} = 1 + x \cdot \frac{1+r}{1+r+m},$$

and since $c_1 \propto W$,

$$c_1 \text{ rises by } x \cdot \frac{1+r}{1+r+m} \%$$

and so does c_2 , by the same percentage. ✓ Verified for $m \in \{0, 0.5, 1, 3\}$ and $r \in \{0.02, 0.05, 0.20\}$ against the closed-form solution.

Equivalently, in elasticity form,

$$\frac{\partial \ln c_1}{\partial \ln y_1} = \frac{y_1}{W} = \frac{1+r}{1+r+m} \quad (6.9.1)$$

Dependence on m . Decreasing, and this is the whole content of the problem. If $m = 0$ — all income is earned today — then the elasticity is 1 and consumption tracks current income one-for-one. As m grows, current income becomes a smaller share of lifetime resources and the same percentage raise moves consumption less and less; as $m \rightarrow \infty$ the elasticity goes to zero.

$m = y_2/y_1$	0	0.5	1	3
elasticity at $r = 0.05$	1.000	0.677	0.512	0.259

Dependence on r . Increasing. A higher interest rate discounts future income more heavily, so period-1 income makes up a larger share of W and the response is stronger. The effect is second-order: going from $r = 0.02$ to $r = 0.20$ at $m = 1$ moves the elasticity only from 0.505 to 0.545.

Reading. Equation (6.9.1) is the permanent income hypothesis in its most compact form, and it is worth committing to memory:

the elasticity of consumption to current income equals current income's share of lifetime wealth.

Three things follow.

- Neither β nor σ appears.** The proportional response is a pure wealth-accounting statement; preferences determine only the *level* of c_1 through Ω , not its elasticity. This makes (6.9.1) unusually robust — it does not require knowing the EIS, the parameter nobody agrees on.
- It predicts who responds to income shocks.** Households with steep expected income profiles (high m : students, residents, young professionals — Problem 6.2) should barely respond to current income; households near the end of their earning life (low m) should respond almost fully.

3. It is the correct way to state “the MPC is small”. The level MPC $\partial c_1 / \partial y_1 = 1/\Omega$ and the elasticity are different objects, and the elasticity is the one that is invariant to units and to preferences.

Interest Rates, Income and Substitution Effects

Problem 6.3 Consumption and Interest Rates

Question

A household solves a special case of the problem in Exercise 6.1, with $\tau_1 = \tau_2 = a_0 = 0$. Suppose that, by coincidence, the values of β , r , y_1 and y_2 are such that it is optimal for the household to consume

$$c_1 = y_1, \quad c_2 = y_2.$$

- What will happen to c_1 if interest rates increase? [A graph will be helpful. Make sure you draw it carefully, it's probably useful to make it large.]
- Does the overall utility achieved by the household increase, decrease or stay the same?
- Suppose this household was the only household in the economy and Jones & Klenow, using this year's data, applied their measure of welfare to this economy. How would their measure of welfare change with the change in interest rates?
- Explain the relationship between the answer to part (b) and the answer to part (c).

Solution:

- c_1 falls, unambiguously, for any $\sigma > 0$.

The graphical argument, which is the one the question wants. The budget line always passes through the endowment point (y_1, y_2) — the household can always choose to consume its income and save nothing — and has slope $-(1+r)$. So a rise in r **rotates the budget line about (y_1, y_2) , making it steeper**. The old bundle is *still exactly feasible*, and it is now an interior point of the budget line with a steeper price line running through it. Since the indifference curve through (y_1, y_2) was tangent to the old, flatter line, at that point the new line is steeper than the indifference curve, so the household can climb to a higher curve by moving **up and to the left**: less c_1 , more c_2 . By strict convexity the new tangency is strictly to the left. See Figure 2.

The algebraic argument. Because $a = 0$ initially, the household is a *net trader of zero*: its wealth, measured at the new prices, is exactly what it was, since it holds

precisely the bundle it is endowed with. In Slutsky terms,

$$\underbrace{\frac{dc_1}{dr}}_{\text{total}} = \underbrace{\frac{\partial c_1}{\partial r} \Big|_{U \text{ const}}}_{\text{substitution} < 0} + \underbrace{(y_1 - c_1) \cdot (\dots)}_{\text{income, } \propto a=0} = \text{substitution effect only} < 0.$$

The income effect is proportional to the household's *net* position, and that position is zero. Hence there is nothing to offset the substitution effect, and c_1 falls regardless of σ — the ambiguity of Problem 6.1(e) is gone precisely because there the household was a net saver.

✓ Numerically: calibrating $\beta = (y_2/y_1)^{-\sigma}/(1+r_0)$ with $y_1 = y_2 = 1$, $\sigma = 2$, $r_0 = 0.05$ puts the household exactly at its endowment ($\beta = 0.952381$). Raising r then gives

r	c_1	c_2	a
0.05	1.00000	1.00000	0.00000
0.08	0.99323	1.00732	0.00677
0.12	0.98477	1.01706	0.01523
0.20	0.96957	1.03651	0.03043

The household becomes a **lender**. That direction is not a coincidence: a net-zero trader always moves to the long side of whichever price has risen.

- (b) **Utility strictly increases.** The old bundle (y_1, y_2) remains feasible after the rotation, so the household can guarantee itself its old utility; it chooses a different bundle, which by revealed preference must be strictly better. This is the standard result that an agent at its endowment point can only gain from a change in relative prices — there is no adverse wealth effect for it to suffer. ✓ The numerical utility gains are +0.00010, +0.00051 and +0.00217 for the three rows above: positive, and *second-order small*, because the household starts at a tangency where the first-order gain from trading is zero.
- (c) **The Jones–Klenow measure falls.** Their measure (§2.2) evaluates living standards from *consumption*, not production — but from consumption *in the year being measured*. This year's consumption is c_1 , and part (a) says c_1 has fallen. Holding leisure, life expectancy and inequality fixed (there is one household, and nothing else in the experiment changed), the measure records a decline. With λ defined by $u(\lambda c^{US}) = u(c^{\text{this economy}})$ and CRRA felicity, measured welfare falls in the same proportion as c_1 .
- (d) **The two answers are both right, and the contradiction is the lesson.** The household is *better off* and *consuming less*, at the same time, because it has chosen to move consumption to the future. The Jones–Klenow number sees the cost of that decision — lower c_1 — and cannot see the benefit, higher c_2 , because the benefit arrives in a year the measure does not look at.

This is a sharper version of the caveat Kurlat attaches to the measure in §2.2. Their headline improvement over GDP is to use *consumption* rather than production, and that is right: production is not welfare. But a single year's consumption is not welfare either, for exactly the same kind of reason. A country that is investing heavily — consuming little now in order to consume much later — will score badly on a one-year consumption measure while being on a path its residents chose and prefer. So will a household saving for retirement, and so will this household after the interest-rate rise.

The general principle. A welfare measure and the decision problem it is meant to evaluate must cover the same horizon. Jones and Klenow are explicit that they also analyse a lifetime version of the experiment, and it is the lifetime version that would get this case right. The single-year version mistakes voluntary intertemporal substitution for immiseration, which is the same error as reading a fall in consumption during an investment boom as a fall in living standards.

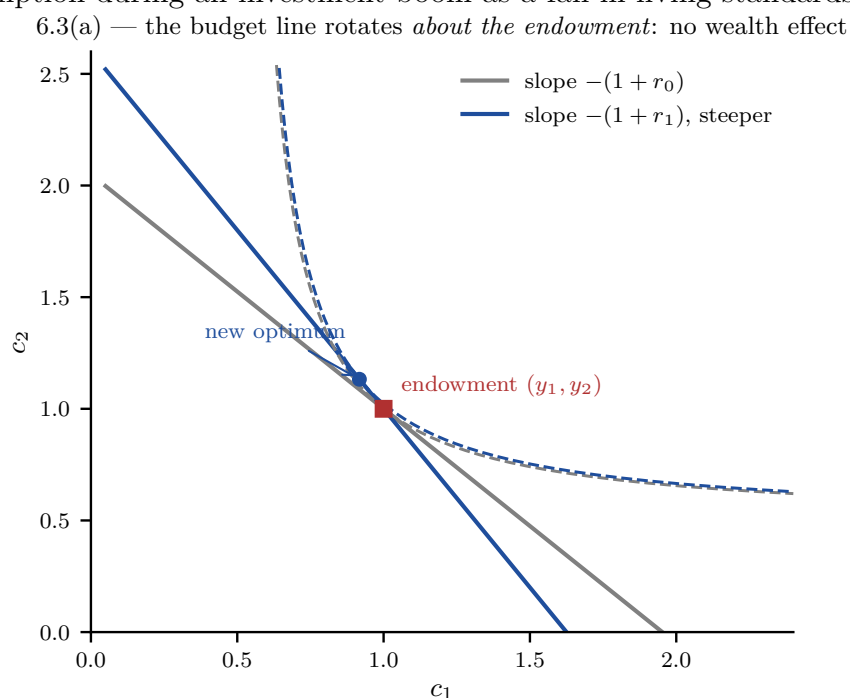


Figure 2: Problem 6.3(a). The budget line rotates about the endowment, so the old bundle stays feasible: pure substitution, c_1 falls, utility rises. Drawn with a large change in r for legibility.

Problem 6.6 A Tax on Savings

Question

A household solves a special case of the problem from Exercise 6.1, with $\tau_1 = \tau_2 = a_0 = 0$. Suppose now that the government introduces a tax on interest income, so that a household that saves a (and therefore earns interest ra) will have to pay τra in taxes. (If the household borrows instead of saving it pays no tax.)

- (a) Plot the new budget constraint.
- (b) Show graphically: (i) an example where the new policy persuades households to save more; (ii) an example where the new policy persuades the household to save less; (iii) an example where the household does not change its decision in response to the new policy. Explain.

Solution:

- (a) The tax falls on interest *income* only, so the household faces **two different interest rates depending on which side of its endowment it lands on**:

$$c_2 = \begin{cases} y_2 + [1 + r(1 - \tau)](y_1 - c_1), & c_1 \leq y_1 \quad (\text{saving, } a \geq 0), \\ y_2 + (1 + r)(y_1 - c_1), & c_1 \geq y_1 \quad (\text{borrowing, } a \leq 0). \end{cases}$$

The budget constraint is therefore **kinked at the endowment point** (y_1, y_2) :

- to the *left* of y_1 (saving) the slope is $-[1 + r(1 - \tau)]$, which is **flatter** than before — each unit of saving now buys less future consumption;
- to the *right* of y_1 (borrowing) the slope is unchanged at $-(1 + r)$.

The endowment point itself is unaffected, since consuming exactly one's income involves no interest income and hence no tax. The absolute slope increases as c_1 increases, so the budget set remains **convex** and the first-order conditions can still be used side by side. See Figure 3.

- (b) Because the policy lowers the return to saving, its effect on the *quantity* saved is exactly the ambiguity of Problem 6.1(e), read in reverse. For a saver, a lower after-tax return means:
 - a **substitution effect** towards present consumption (future consumption is now more expensive): save *less*;
 - an **income effect**: the saver is poorer, so it consumes less of both goods, in particular less today: save *more*.

Which dominates is governed by $EIS = 1/\sigma$. Hence the three cases.

- (i) **The household saves more: a saver with a low EIS (σ large).** Draw sharply curved, almost L-shaped indifference curves. Such a household insists on a particular future standard of living; when the return falls it must put aside *more* to get there. The income effect dominates. ✓ With $y_1 = 2$,

$y_2 = 0.5$, $\beta = 0.97$, $\sigma = 5$, $r = 0.10$ and $\tau = 0.5$, saving rises from $a = 0.72223$ to $a = 0.73398$.

- (ii) **The household saves less: a saver with a high EIS (σ small).** Draw nearly straight indifference curves — the household is close to indifferent about *when* it consumes and reacts strongly to the price. The substitution effect dominates. ✓ Same parameters with $\sigma = 0.4$: saving falls from $a = 0.81371$ to $a = 0.76007$.
- (iii) **The household does not change its decision.** Two clean examples:
- **A borrower.** If the optimum lies to the right of the endowment, it lies on the segment the tax never touched, and nothing at all changes. The tax is on interest *income*, and this household has none. ✓ With $y_1 = 0.5$, $y_2 = 2$, $\sigma = 2$: $a = -0.69552$ before and after, to machine precision.
 - **A household exactly at the kink.** At (y_1, y_2) the budget set has a corner, so a whole interval of marginal rates of substitution, $[1 + r(1 - \tau), 1 + r]$, is consistent with optimality. Any household whose MRS at the endowment falls inside that interval stays put — and the interval is *created* by the policy, so the tax actually makes the no-trade outcome optimal for a positive measure of households that previously saved a little.

Reading, and the quantity response is the wrong thing to look at. The exercise is built to show that the sign of the saving response is not a theoretical result. But three things about the policy are *not* ambiguous, and they are what a policy discussion should turn on:

1. **Every saver is strictly worse off.** Its budget set has shrunk, so utility falls whatever happens to the quantity saved. Case (i), where the household saves more, is not a household that has been made better off — it is one that has been forced to work harder for the same retirement.
2. **The intertemporal margin is distorted.** At the optimum the saver's MRS equals $1 + r(1 - \tau)$ while the economy's marginal rate of transformation is $1 + r$. That wedge is a genuine deadweight loss, and it is present in case (i) exactly as much as in case (ii). This is a First Welfare Theorem failure of the kind catalogued in §9.2.
3. **The asymmetry subsidises debt.** Taxing interest received while leaving interest paid untaxed drives a wedge between the borrowing and lending rates, which is a distortion on top of the level effect — and it is exactly the wedge that makes the no-trade kink attractive in case (iii).

6.6 — a tax on interest income kinks the budget set at the endowment ($r = 0.8$, $\tau = 0.5$, exaggerated for legibility)

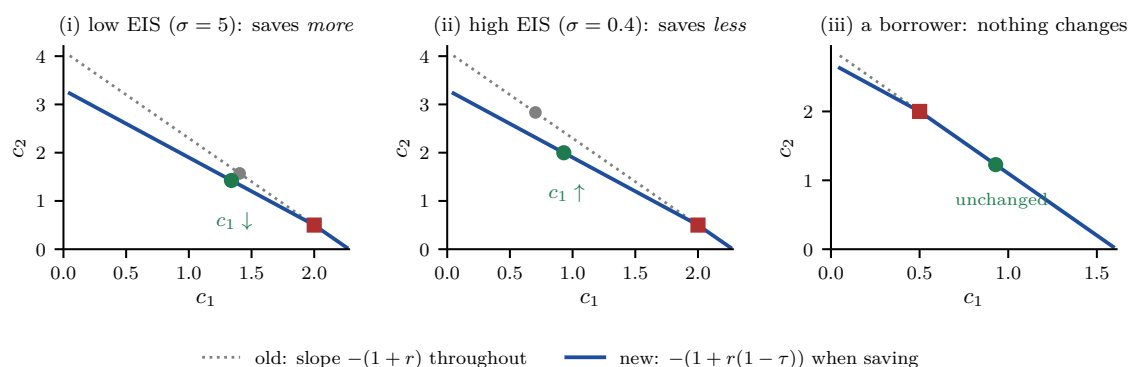


Figure 3: Problem 6.6. The kinked budget set and the three cases. Green dots are the new optima. The interest rate is exaggerated so the kink is visible; the signs are the ones verified numerically at $r = 0.10$.

Permanent Income and the Keynesian Consumption Function

Problem 6.2 Savings Rates for Different People

Question

An economist has data on the occupation, this year's income (denoted y) and this year's consumption (denoted c) of a sample of 26-year-olds. Within this sample, some are top professional athletes and others are medical doctors in their first year of residence. [Make whatever assumptions you think are reasonable and, if you wish, refer back to Exercise 6.1.]

- Suppose one computed $s = \frac{y - c}{y}$ for each individual in the sample. Should we expect s to be higher for athletes or doctors?
- Suppose interest rates go down. How should we expect the response of s to differ between the two groups?

The assumption that does all the work. Both groups are 26 years old, so “this year's income” is y_1 in the language of Problem 6.1 and everything else is y_2 . The two occupations have opposite income profiles:

	Top athlete, 26	First-year resident, 26
Current income y_1	very high (career peak)	very low (a resident's stipend)
Future income y_2	low (careers end near 30)	very high (a physician's career)
$m = y_2/y_1$	small	large

Both are close to the extremes; everything follows from that and from Problem 6.1.

Solution:

- (a) From Problem 6.1 with $\tau = a_0 = 0$,

$$s = 1 - \frac{c_1}{y_1} = 1 - \frac{1}{\Omega} \left(1 + \frac{m}{1+r} \right).$$

This is **decreasing** in m , so

s is higher for the athletes.

✓ With $\beta = 0.97$, $\sigma = 2$, $r = 0.06$: an athlete with $(y_1, y_2) = (10, 2)$ has $s = +0.393$; a resident with $(y_1, y_2) = (3, 9)$ has $s = -0.958$.

The residents' saving rate is **negative**, and this is the right prediction rather than an artefact: first-year residents finance current consumption with student debt. The athletes save heavily because they know their earning years are nearly over. Neither is a statement about thrift; both are the same optimising rule applied to opposite income profiles.

An important corollary for the economist holding the data. A cross-sectional regression of c on y in this sample will find a strong positive relationship and a declining average propensity to consume, and will be tempted to call it a Keynesian consumption function. Problem 6.4 shows exactly how that inference fails.

- (b) Two channels respond to a fall in r , and they are of very different sizes for the two groups.

Channel 1. Human wealth. $W = y_1 + y_2/(1+r)$, so $\frac{dW}{dr} = -y_2/(1+r)^2$: a fall in r raises the present value of future income, *in proportion to* y_2 . This is enormous for the residents, whose income is almost all in the future, and negligible for the athletes.

Channel 2. The price effect. A fall in r changes Ω , tilting the consumption path. Its sign depends on σ (Problem 6.1(e)) and its size does not depend on the income profile.

Channel 1 dominates the comparison and is unambiguous in sign, so

the residents' saving rate falls much more than the athletes'.

✓ Cutting r from 6% to 2%: the athlete's s moves by +0.002, the resident's by -0.038 — about **19 times larger**, and in the opposite direction.

There is also a distributional reading of the same fact. The residents are *borrowers* ($a < 0$) and the athletes are *lenders* ($a > 0$). A fall in r is a transfer from lenders to borrowers, so it raises the residents' consumption and lowers the athletes', on top of the human-wealth channel which points the same way.

Reading. This is the microeconomic foundation for one of the most robust findings in applied macroeconomics: **the interest sensitivity of consumption is concentrated among young borrowers with steep income profiles**. It is why monetary policy transmits partly through household balance sheets rather than uniformly; why the same rate cut has very different effects in a country with a young, indebted population than in one with an old, asset-rich one; and why models that assume a single representative household understate the consumption response to interest rates. Note also that the answer to (b) barely uses σ — the parameter nobody can estimate — and rests almost entirely on the income profile, which is observable.

Problem 6.4 Consumption and Income

Question

Suppose there are two households in the economy. Each of them solves a special case of Exercise 6.1, where $\tau_1 = \tau_2 = a_0 = 0$, $\beta = 1$ and $r = 0$.

- (a) Solve for c_1 as a function of y_1 and y_2 .
 (b) Suppose the incomes of each of the households are:

	y_1	y_2
Household A	2	4
Household B	6	4

Compute c_1 and c_2 for each of the households.

- (c) Suppose an economist is trying to decide what is a reasonable model for consumption behavior and only has data for period 1. Is the data supportive of the Keynesian view of the consumption function? Explain.
 (d) Suppose the same economist now looks at data for period 2 in addition to data for period 1. Is the data supportive of the Keynesian view of the consumption function? Explain.

Solution:

- (a) With $\beta = 1$ and $r = 0$, the Euler equation $c_1^{-\sigma} = \beta(1+r)c_2^{-\sigma}$ collapses to $c_1 = c_2$: **perfect consumption smoothing**, for any σ . Equivalently $\Omega = 1 + 1 = 2$, and the budget constraint is $c_1 + c_2 = y_1 + y_2$, so

$$c_1 = c_2 = \frac{y_1 + y_2}{2}$$

which is average income. Note that neither σ nor anything else about preferences survives: with $\beta(1+r) = 1$ the household simply flattens its consumption path.

(b) Substituting:

	y_1	y_2	c_1	c_2	$a = y_1 - c_1$	$APC_1 = c_1/y_1$
Household A	2	4	3	3	-1 (borrows)	1.50
Household B	6	4	5	5	+1 (saves)	0.83

✓ Each row satisfies $c_1 + c_2 = y_1 + y_2$ and $c_1 = c_2$. Household A has a rising income profile and borrows against it; household B has a falling one and saves.

(c) **Yes — and spectacularly so. That is the trap.** With only period-1 data the economist sees two points, $(y, c) = (2, 3)$ and $(6, 5)$. Fit a line through them:

$$\text{slope} = \frac{5 - 3}{6 - 2} = 0.5, \quad \text{intercept} = 3 - 0.5 \times 2 = 2, \quad \implies \boxed{c = 2 + 0.5y}.$$

✓ This is the Keynesian consumption function $C = \bar{C} + bY$, and it satisfies every one of Keynes's stated properties:

- a positive autonomous term, $\bar{C} = 2 > 0$;
- a marginal propensity to consume strictly between zero and one, $b = 0.5$;
- an average propensity to consume that *falls* with income, 1.50 then 0.83 — the “fundamental psychological law”.

The data were generated by a model in which **current income does not enter the consumption decision at all**, and they fit the Keynesian function perfectly. The reason is that the cross-sectional variation in y_1 here is entirely *transitory*: A and B have the same lifetime income timing but different current income, and the household with temporarily high income is the one with the low APC. That is precisely Friedman's (1957) resolution of the cross-section evidence in Figures 6.1.1–6.1.2.

(d) **No. The second period destroys it.** Pool the four observations:

observation	A, period 1	B, period 1	A, period 2	B, period 2
income y	2	6	4	4
consumption c	3	5	3	5

The last two columns are fatal: **the same income, $y = 4$, is associated with two different consumption levels, 3 and 5.** No function $c = f(y)$ — Keynesian, linear, or otherwise — can fit that, because a function cannot take two values at one point. The Keynesian consumption function is not merely mis-estimated; it does not exist for this data. ✓

What *does* fit is immediate once one looks at it the right way: **within** each household, income changes ($2 \rightarrow 4$ for A, $6 \rightarrow 4$ for B) and consumption does not move at all. Consumption is perfectly correlated with lifetime income across households and perfectly uncorrelated with current income within them.

Where this goes wrong

Do not conclude from (c) that the economist was careless. The period-1 data are real data correctly summarised; the fitted line has the right slope and the right intercept for the sample it was estimated on. The error would be *interpreting* the fitted line as a behavioural relationship — as something that would still hold if you changed a household's current income. It would not: hand household A an extra unit of y_1 and its consumption rises by $1/2$ only because the extra unit is permanent-income-relevant; hand it an extra unit of y_1 offset by a unit less of y_2 and consumption does not move at all.

This is the identification problem in its cleanest form. The cross-sectional slope of 0.5 is not the MPC out of any particular kind of income shock; it is an artefact of *what kind of income variation happens to generate the cross-section*. Change the source of variation and the slope changes. It is the same reason the aggregate time-series APC is roughly constant (Kuznets) while the cross-sectional APC falls with income, and reconciling those two facts is what the permanent income hypothesis was invented to do.

Reading. Three lessons, in increasing order of generality.

1. **A good fit is not evidence for a mechanism.** The Keynesian function fits data generated by a model that contradicts it.
2. **Panel data beat cross-sections** for consumption questions, because they let you hold the household — and hence its lifetime income — fixed while current income moves. This is why the modern literature is built on panel and administrative data.
3. **What kind of income shock is it?** Every question about the marginal propensity to consume is ill-posed until one says whether the shock is temporary or permanent, anticipated or not. Problem 6.8 computes both answers for the infinite-horizon case, and the gap between them is an order of magnitude.

Credit Constraints and Ricardian Equivalence

Problem 6.5 Credit Constraints and Ricardian Equivalence

Question

Suppose a household solves the following variant of the problem in Exercise 6.1:

$$\max_{c_1, a, c_2} u(c_1) + \beta u(c_2) \quad \text{s.t.} \quad a = y_1 - \tau_1 - c_1, \quad c_2 = y_2 - \tau_2 + (1+r)a, \quad a \geq -b \quad (6.4.1)$$

- (a) What does equation (6.4.1) mean? What does b represent?
- (b) Plot the household's budget constraint. In the same graph, plot constraint (6.4.1).
- (c) Solve for c_1 , c_2 and a .
- (d) Show that, other things being equal, constraint (6.4.1) is more likely to be binding if (i) $y_2 - \tau_2$ is high, (ii) $y_1 - \tau_1$ is low, (iii) b is low. Interpret each of these conditions.
- (e) Suppose that the government announces a stimulus package of size Δ . This involves lowering τ_1 by Δ and increasing τ_2 by $\Delta(1+r)$ so that the present value of taxes is unchanged. How does c_1 respond to the stimulus package if we start from a situation where constraint (6.4.1) is NOT binding? How does c_1 respond if we start from a situation where it IS binding? Explain.
- (f) Suppose that instead of announcing a stimulus package, the government announces that it will allow households to borrow Δ from the government and repay it back (with interest) at $t = 2$. How do the effects of this policy compare with the effects of the stimulus package? Explain.

Solution:

- (a) $a \geq -b$ is a **borrowing limit**. Savings a may be negative — the household may borrow — but not below $-b$. So

b is the maximum amount of debt the household may carry into period 2.

$b = 0$ means the household cannot borrow at all; $b = \infty$ recovers Problem 6.1. Note what the constraint is *not*: it is not a solvency condition. The household is perfectly able to repay — it will have $y_2 - \tau_2$ next period — and there is no default in the model. The limit is exogenous, a stand-in for the fact that lenders cannot verify or enforce repayment out of future labour income. That distinction matters for part (f).

- (b) In (c_1, c_2) space:

- the **budget line** is $c_1 + \frac{c_2}{1+r} = W$ with $W = y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}$, a straight line of slope $-(1+r)$ through the endowment $(y_1 - \tau_1, y_2 - \tau_2)$;
- the **credit constraint** is $a = y_1 - \tau_1 - c_1 \geq -b$, i.e.

$$c_1 \leq y_1 - \tau_1 + b,$$

a **vertical line**. The feasible set is the part of the budget line to the left of it. Note the vertical line always lies weakly to the *right* of the endowment (since $b \geq 0$), so the household can always at least consume its own income. See Figure 4.

- (c) Let $c_1^u \equiv W/\Omega$ be the unconstrained solution from Problem 6.1. The objective is strictly concave in c_1 along the budget line, so the constrained maximum is either interior or at the boundary, and

$$c_1 = \min \left\{ \frac{W}{\Omega}, y_1 - \tau_1 + b \right\}$$

$$a = \max \left\{ y_1 - \tau_1 - \frac{W}{\Omega}, -b \right\}, \quad c_2 = y_2 - \tau_2 + (1+r)a$$

Explicitly, in the two regimes:

- **Slack** ($c_1^u \leq y_1 - \tau_1 + b$): $c_1 = W/\Omega$, $c_2 = [\beta(1+r)]^{1/\sigma} W/\Omega$, and the Euler equation holds with equality.
- **Binding** ($c_1^u > y_1 - \tau_1 + b$): $a = -b$, $c_1 = y_1 - \tau_1 + b$, $c_2 = y_2 - \tau_2 - (1+r)b$, and the Euler equation holds as the *inequality* $u'(c_1) > \beta(1+r)u'(c_2)$ — the household would like to move consumption forward and cannot.

That last inequality is the signature of a constrained household and is what the empirical literature looks for.

- (d) The constraint binds iff $W/\Omega > y_1 - \tau_1 + b$. Multiply out and collect:

$$\frac{y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}}{\Omega} > y_1 - \tau_1 + b$$

$$\iff \boxed{\frac{y_2 - \tau_2}{1+r} > (\Omega - 1)(y_1 - \tau_1) + \Omega b} \quad (6.5.1)$$

with $\Omega - 1 = \beta^{1/\sigma}(1+r)^{(1-\sigma)/\sigma} > 0$ and $\Omega > 1$. ✓ Condition (6.5.1) is checked against direct solution of the constrained problem for five different parameter configurations in `ch06_numerico.py`.

Every coefficient in (6.5.1) is positive, so the three claims can be read straight off it:

- (i) $y_2 - \tau_2$ **high** $\Rightarrow$ **binds**. The left-hand side rises. A household that expects to be much richer later *wants* to borrow against that; the more it wants, the more likely the limit stops it. This is the medical resident of Problem 6.2, and it is why credit constraints bite hardest on exactly the households whose lifetime income is highest relative to their current income — students, young professionals, anyone at the start of a steep career.
- (ii) $y_1 - \tau_1$ **low** $\Rightarrow$ **binds**. The right-hand side falls. With little current income, almost all of desired consumption has to be financed by borrowing. Note that

τ_1 enters symmetrically with $-y_1$: *a tax today can itself push a household into the constrained region*, which is the whole point of part (e).

- (iii) **b low $\Rightarrow$ binds.** Mechanical, and it is the policy variable. b is a measure of financial development: how much can a household borrow against future income? Cross-country and cross-time variation in b — credit-card penetration, student-loan availability, mortgage loan-to-value limits — is the standard explanation for why the same income shock produces very different consumption responses in different places and decades.

- (e) **A present-value-neutral tax cut.** The package sets $\tau_1 \rightarrow \tau_1 - \Delta$ and $\tau_2 \rightarrow \tau_2 + \Delta(1+r)$, so the present value of taxes changes by

$$-\Delta + \frac{\Delta(1+r)}{1+r} = 0 \quad \Rightarrow \quad W \text{ is unchanged.}$$

- (i) **Constraint slack: c_1 does not move at all.** $c_1 = W/\Omega$ and W is unchanged, so

$$\boxed{\frac{dc_1}{d\Delta} = 0}.$$

The household saves the entire tax cut — a rises by exactly Δ — in order to pay the higher tax next period. This is **Ricardian equivalence**: the timing of lump-sum taxes is irrelevant, because the household undoes it. ✓

- (ii) **Constraint binding: c_1 rises one-for-one.** Now $c_1 = y_1 - \tau_1 + b$, and τ_1 has fallen by Δ :

$$\boxed{\frac{dc_1}{d\Delta} = 1}.$$

The marginal propensity to consume out of the stimulus is **one**. ✓ Numerically, with $\beta = 0.96$, $\sigma = 2$, $r = 0.05$, $\Delta = 0.2$: an unconstrained household's c_1 stays at 2.02046; a constrained one's rises from 1.05 to 1.25, exactly Δ .

The explanation, and it is the point of the whole problem. Ricardian equivalence is not a claim about people being clever or far-sighted. It is a claim that the tax cut has given the household nothing it did not already have: the ability to move a unit of resources from period 2 to period 1 at the rate $1+r$. An unconstrained household could already do that in the credit market, so the government's offer is redundant and is neutralised. A constrained household could *not* do it, and for that household the tax cut is not a transfer of timing — it is a **relaxation of a binding quantity constraint**, which is why the whole thing is spent.

- (f) **A government loan of Δ is exactly the same policy.** Allowing the household to borrow Δ from the government and repay $(1+r)\Delta$ at $t = 2$ raises the borrowing limit from b to $b + \Delta$, so

$$c_1 = \min \left\{ \frac{W}{\Omega}, y_1 - \tau_1 + b + \Delta \right\}.$$

Therefore:

- if the constraint was slack, **nothing happens**: an extra credit line that the household did not want is worthless;
- if it was binding, c_1 rises by exactly Δ — **identical to the stimulus package**.

✓ Verified numerically: for both household types, c_1 under the loan equals c_1 under the tax cut, to machine precision.

The comparison, and it reframes what fiscal stimulus is. The two policies have the same effect because **they are the same policy**. Cutting τ_1 by Δ and raising τ_2 by $\Delta(1+r)$ *is* lending Δ at rate r and collecting it back. The label “stimulus” is doing no work; the mechanism is credit provision, and it operates only through households whose credit constraint binds.

Two consequences worth drawing out:

- **The government’s comparative advantage is enforcement, not spending.** Private lenders will not lend more than b because they cannot compel repayment out of future income. The government can — through the tax system. That, and not any demand-side story, is what makes the policy work in this model.
- **The loan is better targeted.** The tax cut moves everyone’s taxes, including the unconstrained households who simply save it; the loan is voluntary, so only the households that would respond take it up. In this model the two give the same c_1 , but the loan achieves it without moving the tax bill of anyone who does not need it.

The general moral is the one §9.2 states as a rule: identify which assumption of the First Welfare Theorem fails before recommending a policy. Here the failure is *incomplete markets / borrowing constraints*, and once that is named, the effective policy — relax the constraint — follows immediately, while policies aimed at anything else do nothing.

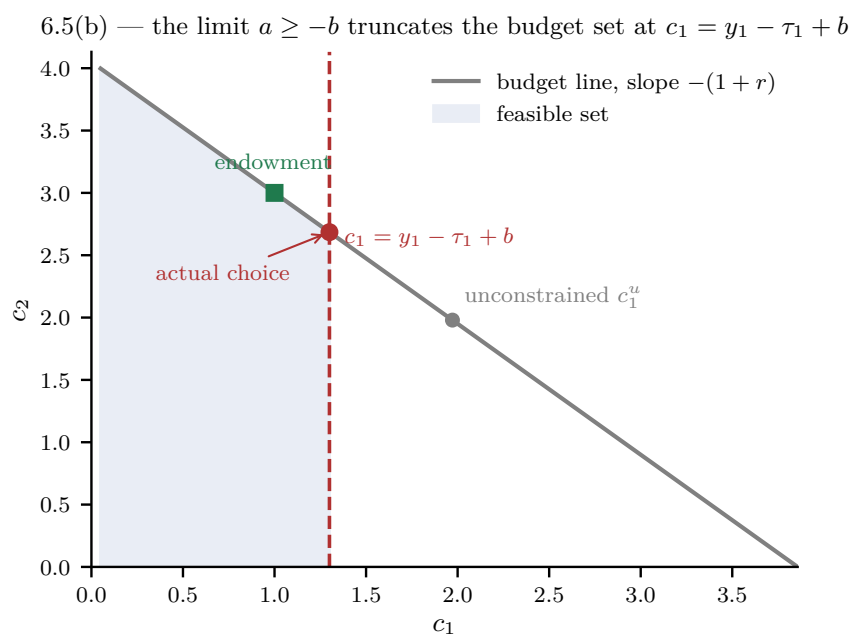


Figure 4: Problem 6.5(b). The borrowing limit truncates the budget set with a vertical line at $c_1 = y_1 - \tau_1 + b$. Here the household would like c_1^u and is pushed back to the boundary.

Many Periods, and Optimal Saving in a Solow Economy

Problem 6.8 The Marginal Propensity to Consume

Question

Suppose a household has preferences given by (6.3.1) over consumption across infinite periods. Its discount factor and the real interest rate satisfy

$$1 + r = \frac{1}{\beta}.$$

The household starts with no wealth, will receive income y_0 in period $t = 0$ and income $\bar{y}$ in every subsequent period, from $t = 1$ onwards.

- (a) Use the Euler equation (6.3.9) and the budget constraint (6.3.8) to find an expression for initial-period consumption c_0 .
- (b) Compute c_0/y_0 .
- (c) Suppose $r = 0.04$. What is the marginal propensity to consume out of a purely temporary increase in income? Describe in words what a household does with a temporary increase in income.

Solution:

- (a) **Step 1: the Euler equation flattens the path.** Equation (6.3.9) is $u'(c_t) = \beta(1+r)u'(c_{t+1})$. With $\beta(1+r) = 1$ this is $u'(c_t) = u'(c_{t+1})$, and since u' is strictly decreasing,

$$c_t = c \quad \text{for every } t.$$

Consumption is *perfectly flat*, whatever σ is and whatever the income path looks like. Note how much this assumption buys: it removes preferences from the problem entirely and leaves pure accounting.

Step 2: write the budget constraint. Equation (6.3.8) with $a_0 = 0$ is

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t}.$$

The left-hand side is a geometric sum with ratio $1/(1+r) < 1$:

$$\sum_{t=0}^{\infty} \frac{c}{(1+r)^t} = \frac{c}{1 - \frac{1}{1+r}} = c \frac{1+r}{r}.$$

The right-hand side splits into the first term and the rest:

$$y_0 + \sum_{t=1}^{\infty} \frac{\bar{y}}{(1+r)^t} = y_0 + \frac{\bar{y}}{1+r} \cdot \frac{1}{1 - \frac{1}{1+r}} = y_0 + \frac{\bar{y}}{r}.$$

Step 3: solve.

$$c \frac{1+r}{r} = y_0 + \frac{\bar{y}}{r} \quad \Rightarrow \quad \boxed{c_0 = c = \frac{r}{1+r} \left(y_0 + \frac{\bar{y}}{r} \right) = \frac{r y_0 + \bar{y}}{1+r}}$$

✓ Verified in `ch06_numerico.py` by simulating 4000 periods of the asset recursion $a_{t+1} = (1+r)a_t + y_t - c_0$ and confirming that the present value of consumption equals the present value of income to 10^{-6} .

Two readings of the same formula, and the second is the useful one. Write the answer as

$$c_0 = \underbrace{\bar{y}}_{\text{permanent income}} + \underbrace{r \cdot \frac{y_0 - \bar{y}}{1+r}}_{\text{interest on the annuitised windfall}}.$$

The household treats the excess $y_0 - \bar{y}$ as a windfall, saves it, and consumes only the interest on it, forever. ✓ Indeed the simulated asset path settles at exactly $a = (y_0 - \bar{y})/(1+r)$ and stays there.

(b) Dividing by y_0 ,

$$\frac{c_0}{y_0} = \frac{r y_0 + \bar{y}}{(1+r) y_0} = \frac{r}{1+r} + \frac{1}{1+r} \cdot \frac{\bar{y}}{y_0}$$

This is a consumption function — but of the *ratio* $\bar{y}/y_0$, not of the level of y_0 . A household with an unusually good year ($y_0 > \bar{y}$) has a low APC and a household with an unusually bad one has an APC above one, which is exactly the cross-sectional pattern Problem 6.4 produces and Keynes misread.

(c) **The marginal propensity to consume out of temporary income.** A “purely temporary” increase raises y_0 and leaves $\bar{y}$ alone:

$$\frac{\partial c_0}{\partial y_0} = \frac{r}{1+r} = \frac{0.04}{1.04} = \boxed{0.0385} \quad \text{i.e. 3.85\%}.$$

✓

In words: the household receives an extra dollar, **saves 96.2 cents of it**, and consumes 3.8 cents. It then goes on consuming an extra 3.8 cents *every year forever*, financed by the interest on the 96.2 cents it put away. It converts the windfall into an annuity and lives off the coupon — because a one-off dollar, spread over an infinite life, is worth only its annuity value.

Contrast with a permanent increase, where y_0 and $\bar{y}$ both rise by one:

$$\frac{\partial c_0}{\partial y_0} + \frac{\partial c_0}{\partial \bar{y}} = \frac{r}{1+r} + \frac{1}{1+r} = \boxed{1}.$$

✓ The permanent MPC is exactly one; the temporary MPC is $r/(1+r)$. At $r = 4\%$ these differ by a **factor of 26**.

Reading — this is both the model’s cleanest prediction and its biggest problem.

- 1. The prediction is unusually sharp.** It contains no preference parameters at all: σ cancelled at the Euler equation, β was pinned by $\beta(1+r) = 1$. The number $r/(1+r)$ is pure arithmetic, which makes it a genuine test of the theory rather than a calibration.

2. **The test fails, and not narrowly.** Measured responses to temporary transfers — tax rebates, stimulus cheques, lottery wins, anticipated tax refunds — come in around 0.2 to 0.6 in the quarter of receipt, five to fifteen times the 0.038 predicted here. The gap is far too large to be measurement error.
3. **The two repairs are the rest of the chapter.** Either the household *cannot* save the windfall's counterpart — it is up against a borrowing constraint and has been consuming less than it wants (Problem 6.5, where the MPC out of exactly this kind of transfer is **one**) — or it does not do the calculation at all (§6.4, behavioural theories: rational inattention, mental accounting, self-control). The modern literature mostly takes the first route, and Problem 6.5(e) is the reason: constraints deliver the observed MPCs without abandoning optimisation.
4. **The infinite horizon is doing real work.** With a finite horizon T the same derivation gives an MPC of roughly $1/T$ rather than $r/(1+r)$ — larger, and larger still for households near the end of life. Part of the measured MPC is therefore just the fact that people do not live forever.

Problem 6.7 One Rational Household

Question

Consider an economy that is well described by the Solow growth model. The production function is $Y = K^\alpha L^{1-\alpha}$. The population is constant and equal to 1 and there is no technological progress. The saving rate is s and the depreciation rate is δ .

- (a) What will the capital stock be in the long run?
- (b) What will be the real interest rate in the long run?

For the rest of the question, assume that $s = 0.4$, $\alpha = 0.35$ and $\delta = 0.1$ and the economy is initially at a steady state.

- (c) If the savings rate increases from $s = 0.4$ to $s = 0.5$: (i) Will GDP increase in the long run? (ii) Will consumption increase in the long run?
- (d) Suppose a single household in the economy (the Friedmans) decides that, instead of just saving an exogenous fraction s of their income, they are going to start choosing consumption and saving to maximize $\sum_{t=0}^{\infty} \beta^t u(c_t)$. The Friedmans can borrow or lend at the market interest rate. Since it's just the Friedmans who start acting this way, and they are small relative to the economy, we are going to assume that the aggregate economy (aggregate quantities, prices, etc.) remains unchanged. Will the Friedmans' consumption be high initially and then fall over time, be low initially and then rise over time, or remain constant?

Solution:

- (a) With $L = 1$, $n = g = 0$, the steady state has $sY = \delta K$:

$$sK^\alpha = \delta K \quad \implies \quad K^* = \left(\frac{s}{\delta}\right)^{\frac{1}{1-\alpha}}, \quad Y^* = \left(\frac{s}{\delta}\right)^{\frac{\alpha}{1-\alpha}}.$$

- (b) $r = F_K - \delta = \alpha K^{\alpha-1} - \delta$. Evaluating at K^* , and using $(K^*)^{\alpha-1} = \delta/s$ from the steady-state condition,

$$r = \alpha \frac{\delta}{s} - \delta = \delta \frac{\alpha - s}{s}$$

✓ Note immediately what this says: **the sign of r is the sign of $\alpha - s$** . The interest rate is negative precisely when the saving rate exceeds the capital share — which is the Golden Rule condition of Problem 5.8, met again from the price side.

- (c) With $\alpha = 0.35$, $\delta = 0.1$:

s	K^*	Y^*	$c^* = (1-s)Y^*$	r
$0.35 = \alpha$	6.8711	1.9632	1.2761	0.0000
0.40	8.4381	2.1095	1.2657	-0.0125
0.50	11.8943	2.3789	1.1894	-0.0300

✓

- (i) **GDP rises: yes.** Y^* is increasing in s always, from 2.1095 to 2.3789, a gain of 12.8%. More saving always buys more capital and more output.
- (ii) **Consumption falls: no.** $c^* = (1-s)(s/\delta)^{\alpha/(1-\alpha)}$ is maximised at the Golden Rule $s_{gr} = \alpha = 0.35$, and the economy is *already past it*. Raising s from 0.4 to 0.5 lowers c^* from 1.2657 to 1.1894, a loss of 6.0%. ✓ A grid search over $s \in (0, 1)$ locates the maximum at $s = \alpha$ to four digits.

This economy is **dynamically inefficient**: it could raise consumption in *every* period, starting immediately, simply by saving less. There is no trade-off to weigh — it is over-accumulating, and the negative interest rate in part (b) is the price signal saying so. Note the contrast with Problem 5.8, where $s = 0.2 < \alpha = 0.6$ and the economy was on the other side.

- (d) **High initially, then falling forever.**

The Friedmans take r as given at the steady-state value from part (b),

$$r = \delta \frac{\alpha - s}{s} = 0.1 \times \frac{0.35 - 0.40}{0.40} = -0.0125,$$

and their Euler equation (6.3.9) with CRRA utility gives the growth factor of their consumption:

$$\frac{c_{t+1}}{c_t} = [\beta(1+r)]^{\frac{1}{\sigma}}.$$

Now $1 + r = 0.9875 < 1$ and $\beta < 1$, so

$$\beta(1 + r) < 1 \quad \implies \quad \boxed{\frac{c_{t+1}}{c_t} < 1 \text{ for every } \beta < 1 \text{ and every } \sigma > 0}$$

✓ Checked for $\beta \in \{0.90, 0.96, 0.99, 0.999\}$ and $\sigma \in \{0.5, 1, 2, 5\}$; every combination gives a factor below one. With $\beta = 0.96$ and $\sigma = 2$ the factor is 0.97365, so the Friedmans' consumption falls by about 2.6% a year.

Why the answer is unambiguous, which it usually is not. Ordinarily the sign of consumption growth is a race between impatience ($\beta < 1$, pushing consumption forward) and the return on saving ($1 + r$, pushing it back). Here the return is *itself* below one: saving physically destroys resources, because the marginal product of capital is less than depreciation. Both forces point the same way, so no assumption about β or σ is needed. The Friedmans consume more than their Solow neighbours today, run their wealth down, and converge geometrically towards zero consumption relative to them.

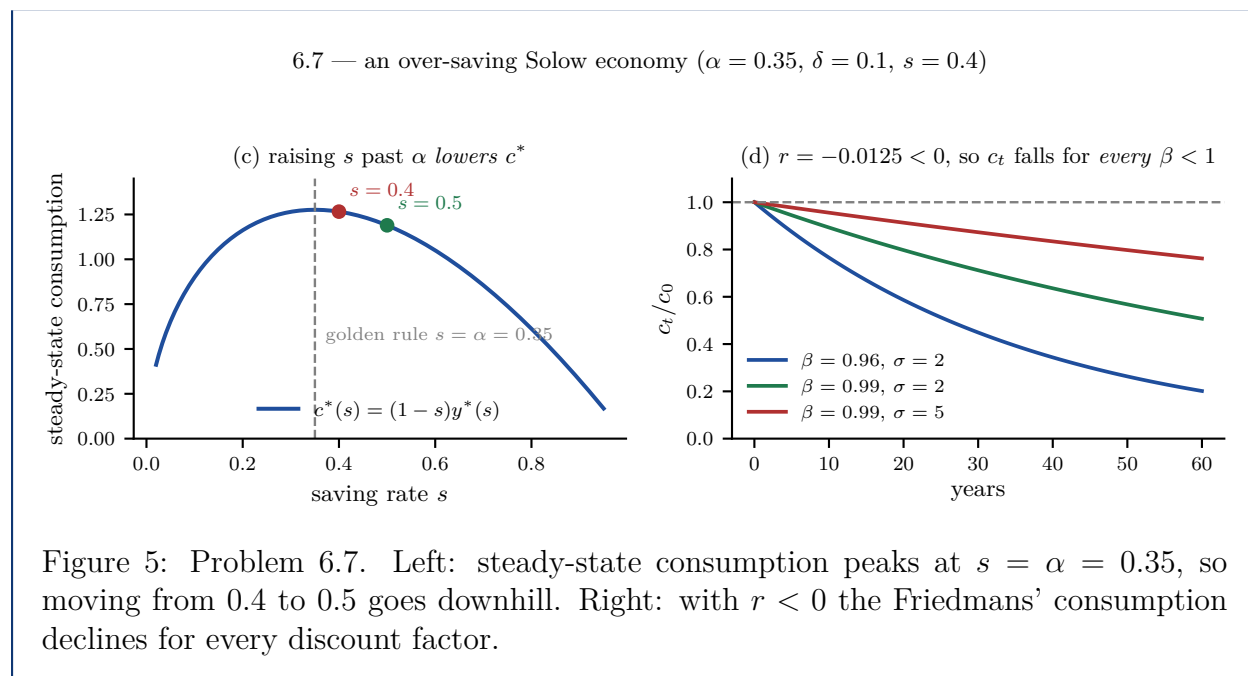
Reading — what this exercise is really demonstrating.

- 1. The exogenous saving rate is a genuinely restrictive assumption, and the model tells you so.** No optimising household would choose $s = 0.4$ here. Given the prices this economy generates, the rational response is to save *less* than everybody else — the Solow assumption is not just arbitrary, it is inconsistent with the behaviour of anyone who looks at the interest rate.
- 2. Part (d) is a partial-equilibrium thought experiment, and it must be.** If all households became Friedmans, aggregate saving would fall, K would fall, F_K would rise and r would rise, until $\beta(1 + r) = 1$ and consumption stopped falling. That is exactly the steady state of Chapter 9:

$$F_K(K^*, 1) - \delta = \frac{1}{\beta} - 1 > 0,$$

which is *strictly below the Golden Rule* and therefore has a strictly positive interest rate. The general-equilibrium version can never be dynamically inefficient; only an exogenous saving rate can produce that. Problem 9.11 works this out.

- 3. The declining path is not a paradox.** It looks strange that a household optimising over an infinite horizon plans for its own consumption to fall forever, but it is the correct response to a negative return: there is no way to store value, so the best plan front-loads. The Friedmans are not making a mistake; the economy they live in is.



Companion code

Every numerical result above is reproduced by `Resolucao/kurlat_ch06_codigo/`. Each script checks its analytical formulas against an independent numerical computation and aborts on any mismatch.

File	What it does
<code>ch06_numerico.py</code>	Problem 6.1: the closed form checked against a direct numerical maximisation of $u(c_1) + \beta u(c_2)$ over the budget line, for four values of σ , plus finite-difference confirmation of all four comparative statics and of the sign rule $\text{sign} \frac{dc_1}{dr} = \text{sign}(\sigma - 1)$. Problem 6.9: homotheticity and the elasticity y_1/W , over a grid in m and r . Problem 6.3: β calibrated so the household sits exactly at its endowment, then $c_1 \downarrow$, $a > 0$ and $U \uparrow$ verified. Problem 6.6: the kinked budget set solved by comparing both segments, producing the three cases. Problem 6.2: the two income profiles, their saving rates and their responses to a rate cut. Problem 6.4: perfect smoothing, the Keynesian fit $c = 2 + 0.5y$, and the pooled data that admit no consumption function. Problem 6.5: the closed-form binding condition (6.5.1) checked against direct solution, the three monotonicity claims, and the equivalence of the tax cut and the government loan. Problem 6.8: a 4000-period simulation confirming the flat path, the annuitised windfall and both MPCs. Problem 6.7: the steady state, the golden-rule grid search, and the Euler growth factor for a grid of β and σ .
<code>ch06_figuras.py</code>	The five figures.
<code>estilo_mpl.py</code>	pgf backend, so figure text is typeset by <code>pdflatex</code> with <code>lmodern</code> and <code>cmap</code> and the PDF stays fully copyable.